

MUSTAFA MOHAMED HAMDY

AI Automation Engineer

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PROFESSIONAL SUMMARY

AI Automation Engineer who designs, ships, and operates production-grade automation and AI systems end-to-end. Built **40+ production n8n workflows** integrating **LLMs (AWS Bedrock, Azure OpenAI GPT-4o), Google Vision OCR, REST APIs / webhooks**, and **20+ SaaS platforms** to automate lead generation, CRM data pipelines, and executive reporting. Delivered an AI LinkedIn engagement engine across **400+ profiles**, AI contact segmentation across **25,000+ records**, and OCR-driven data pipelines — reducing manual workload by **60–90%**. Strong foundation in Python, JavaScript, and SQL, with a focus on reliability: retries, rate limiting, validation, monitoring, and human-in-the-loop.

CORE SKILLS

Automation & Orchestration: n8n (Advanced, self-hosted), Workflow Automation, Business Process Automation, UiPath, Power Automate, Webhooks, Cron/Event Triggers

AI & LLMs: AWS Bedrock, Azure OpenAI (GPT-4o), LLM Integration, AI Agents, Prompt Engineering, Structured Outputs, LLM Classification, AI-based OCR (Google Vision), Human-in-the-Loop, Tool/Function Calling

Programming: Python, JavaScript (Node.js), SQL, Shell Scripting, JSON / Schema

Integration & APIs: REST APIs, OAuth2, Webhooks, ETL Pipelines, HubSpot, Brevo, Jira, Gmail / Google Workspace, WhatsApp Business API, Instagram Graph API, LinkedIn API, Unipile, Apify, Supabase, Mixpanel, Microsoft Clarity, Firebase, NeverBounce, Clay

Cloud & Reliability: AWS, Google Cloud, Firebase, VPS / Self-Hosting, Error Handling, Retry Logic, Rate Limiting, Data Validation, Deduplication, Idempotency, Drift Detection, Monitoring, Structured Logging

PROFESSIONAL EXPERIENCE

AI Automation Engineer — FortyGuard (USA, Remote)

2025 – Present

- Architected FortyGuard's full automation platform on **self-hosted n8n** — **40+ production workflows** integrating AWS Bedrock, Azure OpenAI (GPT-4o), Google Vision OCR, HubSpot, Jira, Gmail, WhatsApp Business, LinkedIn, and Apify — reducing manual operational overhead by **~60%**.
- Built an **AI LinkedIn engagement engine** orchestrating **400+ target profiles** across **5 scheduled daily runs**, generating tone-matched AI comments (GPT-4o / AWS Bedrock) with behavioral pacing, reply detection, and contextual follow-ups.
- Engineered an **AI DM outreach agent** (Unipile API + AWS Bedrock) generating founder-style personalized messages for HIGH-qualified leads, syncing status to Google Sheets.
- Developed a **Commercial Pipeline Extraction** system that mines shared inboxes, applies heuristic + AI qualification to surface B2B opportunities, auto-drafts on-thread follow-ups, and alerts leadership to stale deals (>7 days) via WhatsApp (**human-in-the-loop**).
- Built an **AI contact-segmentation pipeline** classifying **25,000+ HubSpot contacts** into 6 market categories via email-domain grouping + website-content LLM classification, auto-populating campaign lists.
- Designed an **OCR → CRM pipeline** (WhatsApp Business API → Google Vision OCR → LLM structured extraction → HubSpot upsert with deduplication) converting business-card photos into structured contacts in seconds.

- Enriched **4,740 LinkedIn profiles** via batched Apify actors (500/batch), cutting API executions from 4,740 to 10; ran a US ClimateTech campaign (Clay) producing **785 verified senior-contact emails** from 2,000+ companies.
- Authored automated **executive reporting** (Monday Briefing, Midweek Drift, Friday Debrief, Jira↔Email Reconciliation) using multi-stage AI extraction, drift classification, and **RAG (Red/Amber/Green) status**, delivering HTML reports to leadership.
- Implemented **retries, rate-limit control, validation, deduplication/idempotency, structured logging, and monitoring** across all production workflows.

AI Automation Engineer — Elex (Saudi Arabia)

2024 – 2025

- Designed scheduled, **alert-driven data automation pipelines** connecting internal systems with the corporate website, replacing recurring manual database operations.
- Built **SQL- and script-based automations** for backup, validation, integrity checks, and recovery, with real-time monitoring on critical data flows.
- Developed scripts to streamline data movement between web and internal applications, improving cross-system consistency and reducing operations response time.

IT Specialist — Sama International College (Egypt)

2024

- Supported **50+ daily users** across hardware, software, and network infrastructure.
- Introduced preventive-maintenance protocols and **Python automation scripts**, reducing system downtime by **25%**.

KEY PROJECTS

AI LinkedIn Lead Engagement Engine · n8n · AWS Bedrock · GPT-4o · Apify · Unipile · Google Sheets

Multi-workflow system: 5 parallel scraping engines → AI comment generation → posting → reply detection → AI DM outreach for qualified leads. **400+ profiles, 5 runs/day.**

WhatsApp → CRM Business-Card Pipeline · n8n · Google Vision OCR · LLM · HubSpot · Google Drive

Converts business-card photos sent over WhatsApp into deduplicated, structured HubSpot contacts in seconds via OCR + LLM extraction.

AI Contact Segmentation (HubSpot) · n8n · Azure OpenAI · Domain Enrichment

Classifies **25,000+ contacts** into 6 market segments via domain grouping + website-content LLM classification; auto-populates active lists.

Executive Reporting & Drift-Detection Suite · n8n · Jira · AI Extraction · RAG Status

7+ interlinked workflows ingesting briefings/debriefs, reconciling Jira vs email commitments, and detecting scope/risk drift for two teams — delivered as automated HTML reports.

EDUCATION

B.Sc. Computer Science & Information Systems — October 6 University, Egypt · 2024

LANGUAGES

English: Professional Working Proficiency · Arabic: Native